

Rachel Hoffman-Bice

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Machine Design and Robotics Engineer seeking full time position focusing on the development, implementation and production-readiness of control systems, electro-mechanical interfaces and mechanical system design and development.

SKILLSET: Creo, Solidworks, MATLAB, Python, Git, Simulink, ROS, GD&T ASME Y14.5, IEC 60601, IEC 62304, ISO 14971 SAFe Agile, Jira, Raspberry Pi, OpenCV, Injection Molding, Rapid Prototyping, PTC Windchill, DFM/DFA, PPAP

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

Ph.D. in Mechanical Engineering (Machine Design and Robotics), Minor in Leadership, GPA: 4.9/5.0

June 2021

S.M in Mechanical Engineering, GPA: 4.9/5.0

June 2017

University of Florida, Gainesville, Florida, GPA: 3.97/4.0

May 2015

B.S. in Mechanical Engineering, Minor in Electrical Engineering, *summa cum laude*, *Outstanding Scholar Award*

PROFESSIONAL EXPERIENCE

Medtronic Surgical Robotics – Robotic Arm Hardware and Software Subject Matter Expert

February 2022 -Present

Senior Mechanical Design Engineer – New Product Development

May 2024 -Present

- Serve as Robotic Arm and Setup Arm (RASA) Hardware and Software Subject Matter Expert across the Surgical Robotics organization, leading design decisions, field failure investigations, testing and validation of both subsystems, with particular focus as hardware SME to multiple Hugo software release programs
- Technical lead for the next generation Control Arm (Insertion Axis (IA) and Instrument Drive Unit (IDU))—owned requirements definition, technical drawings per ASME Y14.5 and DFMEA-driven mechanical design including tolerance stack analysis, FEA, and validation against IEC 60601 and ANSI/AAMI HE75 requirements. Designed and prototyped three critical subsystems (actuator, pin brake, emergency release) from concept through mechanical validation using Creo, with design iterations based on manufacturing feedback; worked with machine shops on CNC optimization to advance designs toward production readiness
- Technical Lead (Mechanical and Software) of a Robotic Arm actuator field brake failure Corrective and Preventative Action, setting the investigation strategy and authoring comprehensive engineering reports including failure cause mapping, component change impact assessments, vendor testing, and failure cause map analysis; closed with design changes released to production
- Designed for Manufacturability across several manufacturing modalities: CNC milling (actuator housings, pin brake mechanism), injection molding (Control Arm cover), sheet metal fabrication (actuator test rig) and 3D printing (functional prototypes); collaborated with suppliers on DFM reviews, tolerance stackups and design iteration cycles to validate manufacturability before production commitment
- Coordinated across electrical, operations, supply chain, human factors and product leadership teams to define Control Arm subsystem requirements and resolve design tradeoffs; represented the Control Arm in cross-functional design reviews; incorporated feedback from field studies into design improvements; managed supplier relationships for component sourcing and testing
- On a prior next generation platform, held technical ownership of the Control Arm primary actuator, emergency release, custom brake development and navigation hardware to IEC 60601 requirements, from concept to prototype delivery
- Supported next generation platform user studies in Boston and London as the hardware and software subject matter expert, including prototype preparation, on-site troubleshooting, shipment coordination and remote technical support, and summarized clinician feedback into design improvements
- Completed specialized training in GD&T Fundamentals, IEC-60601 medical device compliance, IEC-62304 software lifecycle, ISO-14971 risk management, PTC Creo Fundamentals and DRM (Six-Sigma) to enhance technical and leadership capabilities

Senior Controls Software Engineer

January 2023-May 2024

Controls Software Engineer II

February 2022-January 2023

- Serve as Robotic Arm and Setup Arm (RASA) software technical expert, developer and Product Owner for the Surgical Robotics organization, developed code for 37 functional software modules, authored and maintained documentation for the design, testing and validation required for the RASA software release, including, 23 requirements and 18 design documents released into Agile, and led the team through maintenance and execution and debugging of 77 verification test cases
- Developed Simulink software for control strategies, safety workflows, joint validation functions and operating-mode updates (Manual Control, Teleoperation, Initialization) for a robotic arm with 6 actuated joints (one linear), 4 passive setup joints and a 4-DOF Instrument Drive Unit
- Partnered with quality engineers to derive the DFMECA (per IEC 62304 and ISO 14971) and confirm software mitigations for identified risks were implemented, tested and captured in the design history file
- Integrated a Wristed Ligasure instrument plant model into a matlab simulator to enable Model-in-the-Loop testing without hardware, and used system identification to develop a custom jaw controller
- Re-architected the Instrument Drive Unit (IDU) controller for plug-and-play instrument support, incorporating software architecture team feedback and delivering a prototype that built and deployed to a complete robotic system within one Program Increment, then debugged it to full teleoperation with instrument exchanges before handing it off to the next development team

Medtronic GYN Minimally Invasive Therapies Group, Engineering Contingent

June 2021-February 2022

- Extensive usage of Solidworks to develop ASME Y14.5 compliant assembly drawings for export to suppliers for manufacturing
- Conducted verification testing on minimally invasive hysteroscopes to ensure resection performance met design specifications

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MMID Full Service Design Team, Delft, The Netherlands, *Functionality Design Intern*

May 2017-Aug-2017

- Extensive concept development of a high-pressure pipe coupling, an apparatus for testing chemical dilution and a versatile electric outlet wall box, and a wall mounted broom clamp

MIT Lincoln Laboratory, Systems and Analysis Group, *Mechanical Engineering Intern*

May 2014-Aug-2014

- Evaluated and tested strategies to develop realistic optimal DTED terrain following aircraft trajectories with the goal of minimizing flight time and subject to velocity and acceleration constraints using the MATLAB optimization toolbox

RESEARCH

d'Arbeloff Laboratory, Massachusetts Institute of Technology

August 2015-June 2021

Ph.D Research

- Investigated methods by which heavy shafts suspended with multiple cables at specific positions and orientations can be inserted into a chamfered hole despite a $<200\mu\text{m}$ clearance, resulting in a four-bar-linkage model that can predict successful insertion
- Analyzed system force balance to determine the required balance of cable tension to prevent the peg from jamming during the insertion process and to correct for wedging using cables
- Designed a prototype of a three-winch system with linear actuators and brushless motors designed to manipulate a 15 kg payload

Masters Research

- Invented a novel mechanism by which elevators can travel along tracks, and transfer between track segments of varying direction. This mechanism was inspired by a railway turnabout and the rack-pinion drive system of cog-railways.
- Utilized traditional shortest-path finding algorithms to analyze optimal scheduling routes for multiple carts using this system and detailed the advantages over current elevator systems in e-commerce fulfillment centers.
- Designed and manufactured/implemented scale model consisting of thirteen direction changing mechanisms and twenty-seven track segments, including a Solidworks model, 3D printed and hand-machined components

SELECT PUBLICATIONS (Total: 4 Journal and 2 Conference publications)

- **R. Hoffman** and H. H. Asada, "*Precision Assembly of Heavy Objects Suspended With Multiple Cables From a Crane*," in IEEE Robotics and Automation Letters, vol. 5, no. 4, pp. 6876-6883, Oct. 2020, doi: 10.1109/LRA.2020.3013845
- **Hoffman, R.**, and Asada, H., "A Multi-Track Elevator System for E-Commerce Fulfillment Centers", *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Vancouver, 2017. **Best Application Paper Award**